<u>Last time</u>: For any sequence $\{a_n\}_{n\geq 1}$,

$$\limsup_{n\to\infty} a_n = \lim_{N\to\infty} \underbrace{\sup_{n\geq N} a_n}_{= v_N,\ \text{increasing}}, \qquad \liminf_{n\to\infty} a_n = \lim_{N\to\infty} \underbrace{\inf_{n\geq N} a_n}_{= u_N\ \text{decreasing}}$$

exist in $\mathbb{R}\cup\{\pm\infty\}$.

<u>Thm</u> Let $\{a_n\}_{n\geq 1}$ be a sequence.

① If $\lim_{n\to\infty} a_n$ exists in $\mathbb{R}\cup\{\pm\infty\}$, then

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n.$$

② If $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$ in $\mathbb{R}\cup\{\pm\infty\}$, then

$$\lim_{n\to\infty} a_n = \liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n.$$

- Here, "$\lim_{n\to\infty} a_n$ exists in $\mathbb{R}\cup\{\pm\infty\}$" just means "$\{a_n\}_{n\geq 1}$ converges, or diverges to $+\infty$, or diverges to $-\infty$".

<u>Pf</u>: ① Last time.

② Suppose $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$. Want: $\lim_{n\to\infty} a_n$ exists.

There are 3 cases.

Case 1: $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = -\infty$. Fix $M<0$.

Then $\exists N_0 \in \mathbb{N}$ s.t.

$$v_N < M \quad \forall N \geq N_0$$

$$\Rightarrow \quad v_{N_0} < M$$

$$\Rightarrow \quad a_n \le \sup\{a_n : n \ge N_0\} < M \quad \forall n \ge N_0$$

So $\lim_{n\to\infty} a_n = -\infty$.

Case 2: $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = +\infty$. Similar. (Exercise.)

Case 3: $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = a \in \mathbb{R}$.

Fix $\varepsilon > 0$. As $a = \liminf_{n\to\infty} a_n = \lim_{N\to\infty} u_N$, $\exists N_1 \in \mathbb{N}$ s.t.

$$|u_N - a| < \varepsilon \quad \forall N \ge N_1$$

$$\Rightarrow \quad a - \varepsilon < u_{N_1} < a + \varepsilon$$

$$\Rightarrow \quad a - \varepsilon < \inf\{a_n : n \ge N_1\} \le a_n \quad \forall n \ge N_1$$

As $a = \limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N$, $\exists N_2 \in \mathbb{N}$ s.t.

$$|v_N - a| < \varepsilon \quad \forall N \ge N_2$$

$$\Rightarrow \quad a - \varepsilon < v_{N_2} < a + \varepsilon$$

$$\Rightarrow \quad a_n \le \sup\{a_n : n \ge N_2\} < a + \varepsilon \quad \forall n \ge N_2$$

Together, we get

$$a - \varepsilon < a_n < a + \varepsilon \quad \forall n \ge \max\{N_1, N_2\}$$

$$\Rightarrow \quad |a_n - a| < \varepsilon \quad \forall n \ge \max\{N_1, N_2\}$$

So $\lim_{n\to\infty} a_n = a$. $\square$

<u>Thm</u> (Cesàro–Stolz) Let $\{a_n\}_{n\ge1}$ be a sequence of nonzero real numbers. Then

$$\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| \le \liminf_{n\to\infty} |a_n|^{\frac{1}{n}} \le \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} \le \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|.$$

<u>Ex</u> Find $\lim_{n\to\infty} \sqrt[n]{n}$.

Setting $a_n = n$, we have

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{n+1}{n} \longrightarrow 1 \quad \text{as} \quad n\to\infty$$

$$\Rightarrow \liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1 = \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

So, by Cesàro-Stolz,

$$\liminf_{n\to\infty} |a_n|^{\frac{1}{n}} = 1 = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}$$

$$\Rightarrow \lim_{n\to\infty} \underbrace{|a_n|^{\frac{1}{n}}}_{= n^{\frac{1}{n}} = \sqrt[n]{n}} = 1$$

$\square$

<u>Ex</u> Find $\lim_{n\to\infty} \sqrt[n]{n!}$.

Setting $a_n = n!$, we have

$$\left|\frac{a_{n+1}}{a_n}\right| = n+1 \longrightarrow +\infty \quad \text{as} \quad n\to\infty$$

So, by Cesàro-Stolz,

$$\lim_{n\to\infty} \underbrace{|a_n|^{1/n}}_{(n!)^{1/n} = \sqrt[n]{n!}} = +\infty$$

$\square$

<u>Ex</u> Find $\lim_{n\to\infty} \frac{1}{n}\sqrt[n]{n!}$.

Note $\frac{1}{n}\sqrt[n]{n!} = \left(\frac{n!}{n^n}\right)^{\frac{1}{n}}$. Setting $a_n = \frac{n!}{n^n}$,

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{n^n \cdot (n+1)!}{(n+1)^{n+1} n!} = \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+\frac{1}{n})^n} \to \frac{1}{e} \text{ as } n \to \infty$$

So, by Cesàro-Stolz,

$$\lim_{n\to\infty} \underbrace{|a_n|^{1/n}}_{\left(\frac{n!}{n^n}\right)^{\frac{1}{n}} = \frac{1}{n}\sqrt[n]{n!}} = \frac{1}{e}.$$

$\square$

<u>Lem</u> If $r>0$, then $\lim_{n\to\infty} r^{\frac{1}{n}} = 1$.

<u>Pf</u>: Case: $r \geq 1$. Using the identity

$$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$$

with $a = r^{\frac{1}{n}}$ and $b = 1$, we get

$$r - 1 = (r^{\frac{1}{n}} - 1)\left(\underbrace{r^{\frac{n-1}{n}}}_{\geq 1} + \underbrace{r^{\frac{n-2}{n}}}_{\geq 1} + \dots + \underbrace{r^{\frac{1}{n}}}_{\geq 1} + 1\right)$$

$$\geq (r^{\frac{1}{n}} - 1) \cdot n$$

$$\Rightarrow 0 \leq r^{\frac{1}{n}} - 1 \leq \frac{r-1}{n} \to 0 \text{ as } n \to \infty$$

So $\lim_{n\to\infty}(r^{\frac{1}{n}} - 1) = 0$ by the squeeze theorem (HW4).

Case: $0 < r < 1$. Then $\frac{1}{r} > 1$, so by the previous case,

$$\lim_{n\to\infty} \left(\frac{1}{r}\right)^{\frac{1}{n}} = 1$$

This is nonzero, so by the limit law for division we have

$$\lim_{n\to\infty} \underbrace{\frac{1}{\left(\frac{1}{r}\right)^{\frac{1}{n}}}}_{= r^{\frac{1}{n}}} = 1.$$

$\square$

<u>Thm</u> (Cesàro–Stolz) Let $\{a_n\}_{n\geq 1}$ be a sequence of nonzero real numbers. Then

$$\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| \underset{①}{\leq} \liminf_{n\to\infty} |a_n|^{\frac{1}{n}} \underset{②}{\leq} \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} \underset{③}{\leq} \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|.$$

<u>Pf</u>: ② Immediate.

③ Set $\ell = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} \geq 0$

$L = \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| \geq 0$

Want: $\ell \leq L$. If $L = +\infty$ then this is automatically true, so assume $L \in [0, \infty)$.

It suffices to prove $\ell \leq M$ $\forall M > L$. Fix $M > L$.

$$M > L = \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{N\to\infty} \sup_{n\geq N} \left|\frac{a_{n+1}}{a_n}\right|$$

HW4#4

$$\Longrightarrow \quad \exists N_0 \in \mathbb{N} \text{ s.t. } \sup_{n\geq N} \left|\frac{a_{n+1}}{a_n}\right| < M \quad \forall N \geq N_0$$

$$\Longrightarrow \quad \left|\frac{a_{n+1}}{a_n}\right| \leq \sup_{n\geq N_0} \left|\frac{a_{n+1}}{a_n}\right| < M \quad \forall n \geq N_0$$

$$\Longrightarrow \quad |a_{n+1}| < M|a_n| \quad \forall n \geq N_0$$

By induction,

$$|a_n| < M|a_{n-1}| < M^2|a_{n-2}| < \dots < M^{n-N_0}|a_{N_0}| \quad \forall n > N_0$$

$$\Longrightarrow \quad |a_n|^{\frac{1}{n}} < \left(M^{n-N_0}|a_{N_0}|\right)^{\frac{1}{n}} = M \underbrace{\left(\frac{|a_{N_0}|}{M^{N_0}}\right)^{\frac{1}{n}}}_{\to 1 \text{ as } n\to\infty \text{ by previous lemma}} \quad \forall n > N_0$$

HW4#3

$$\Longrightarrow \quad \ell = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{N\to\infty} \sup_{n\geq N} |a_n|^{\frac{1}{n}} \leq M$$

As $M > L$ was arbitrary, we conclude $\ell \leq L$.

(1) Similar. (Exc) $\square$